Name: ____________________

Class: ____________________

Date: ____________________

SHM 1 – Period of a Spring Mass Oscillation

Practical

A great deal of physics is oscillation: a mass on a spring, a pendulum, a guitar string, a molecule in the atmosphere absorbing infrared, an atom in a crystal. They look nothing like each other. They obey the same equation.

Today you meet the oscillator that is easiest to hold in your hand, and you measure the one thing about it that is genuinely surprising.

IB Physics Guide • §C.1 Simple harmonic motion • *Theme C – Wave Behaviour*

Reproduced from the *IB Physics guide*, first assessment 2025. This is everything C.1 asks of you. We will not do all of it today — but you should be able to see, at any point in the topic, which part of this list you are standing on.

Understandings — standard level and higher level: 3 hours

Students should understand:

- ☐ conditions that lead to simple harmonic motion
- ☐ the defining equation of simple harmonic motion as given by $a = -\omega^2 x$
- ☐ a particle undergoing simple harmonic motion can be described using time period T, frequency f, angular frequency ω, amplitude, equilibrium position, and displacement
- ☐ the time period in terms of frequency of oscillation and angular frequency as given by $T = \dfrac{1}{f} = \dfrac{2\pi}{\omega}$
- ☐ the time period of a mass–spring system as given by $T = 2\pi\sqrt{\dfrac{m}{k}}$
- ☐ the time period of a simple pendulum as given by $T = 2\pi\sqrt{\dfrac{l}{g}}$
- ☐ a qualitative approach to energy changes during one cycle of an oscillation.

Additional higher level: 4 hours

Students should understand:

- ☐ that a particle undergoing simple harmonic motion can be described using phase angle
- ☐ that problems can be solved using the equations for simple harmonic motion as given by

$$x = x_0 \sin(\omega t + \phi) \qquad v = \omega x_0 \cos(\omega t + \phi) \qquad v = \pm\,\omega\sqrt{x_0^2 - x^2}$$
$$E_T = \tfrac{1}{2} m\omega^2 x_0^2 \qquad E_p = \tfrac{1}{2} m\omega^2 x^2.$$

Guidance

- ☐ The significance of the minus sign in the defining equation for simple harmonic motion should be understood.
- ☐ Energy changes during simple harmonic motion (kinetic, potential and total) should be described qualitatively.
- ☐ A quantitative approach to energy changes during simple harmonic motion is required at higher level only.
- ☐ Radians are used for phase angle calculations.

Where today sits

Today answers the **first**, **third** and **fifth** understandings — the conditions for SHM, the quantities that describe an oscillator, and the time period of a mass–spring system — and begins the fourth. The higher level box later in this packet reaches forward into the equations above. Everything else comes next lesson and beyond.

Name: ______________________________ **Date:** ______________________

What makes an oscillation *simple harmonic*

Key concept

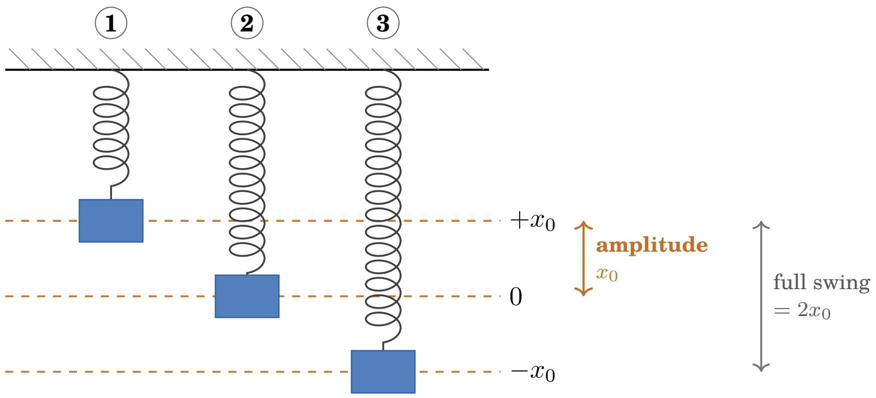
Not every repeating motion is simple harmonic motion. SHM is the special case in which **two conditions** hold:

1. the acceleration is **proportional to the displacement** from the equilibrium position, and
2. the acceleration is always directed **towards the equilibrium position** — that is, opposite to the displacement.

Written as one statement, $a \propto -x$. You will meet the full form $a = -\omega^2 x$ next lesson; for today, what matters is the **minus sign**. It is not decoration. It says the acceleration always points back the way the object came, which is exactly why the motion repeats instead of running away.

The words you need today

Every word below is a label on this picture. It is the same spring and the same mass, drawn at three moments of one oscillation.



1 highest point **2** equilibrium — where it hangs at rest **3** lowest point

The mass is momentarily *at rest* at **1** and **3**, and moving *fastest* at **2**. Note that the amplitude is the distance from the middle to one end — not end to end.

- **Equilibrium position** — where the mass sits at rest, with the spring already stretched by its weight. All displacements are measured from here, not from the unstretched spring.
- **Displacement** x — how far the mass is from equilibrium, with a sign for direction.
- **Amplitude** x_0 — the largest displacement reached. A distance, so always positive.
- **Period** T — the time for *one complete* oscillation: down, back up through equilibrium, and back to where it started, moving the same way.
- **Frequency** f — oscillations per second, in hertz. Period and frequency are the same fact stated two ways:

$$T = \frac{1}{f}$$

The prediction you are testing

Key concept

For a mass m hanging on a spring of spring constant k, theory predicts

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Notice what is *not* in that expression: the amplitude. The prediction says a large swing and a small swing take the **same time**. Most people find that wrong when they first hear it. You are about to check it.

Why we are not deriving this today

Deriving that equation needs Newton's second law and the force law for a spring — both of which you meet with your other physics teacher later this term, in Theme A.
That is deliberate, and it is the honest order for an experimental subject: **today you establish by measurement that the relationship is true, and later you find out why it has to be.** A result you have measured yourself is a much better thing to attach a derivation to than a formula you were handed.

Apparatus and safety

You will need

- Spring, stand, boss and clamp
- Slotted masses and hanger
- Stopwatch
- Metre stick *(make sure yours has millimetre markings, not just centimetres)*
- Digital balance — for weighing your load

Before you start

- Clamp the stand to the bench, or weight the base.
- Keep the masses over the bench, never over a foot.
- Do not overload the spring. If it does not return to its original length when unloaded, you have gone too far — tell me and take a new one.

Method

1. Hang the spring from the clamp and add the hanger. Add masses until the spring is comfortably stretched but well short of its limit. This is your smallest load — record it.
2. Let the mass settle. Where it hangs at rest is the **equilibrium position**.
3. Pull the mass down a small distance — about 2 cm — and release it *without* pushing. The motion should be straight up and down. If it swings sideways or twists, stop it and start again.
4. Time **10 complete oscillations**. Start counting from zero as you start the watch.
5. Time it **three times** for the same mass. Three is not fussiness — it is the smallest number that lets you work out an uncertainty at all, which is the next section.
6. Add mass and repeat. Aim for **six different masses**.

Why 10 oscillations and not one

Your reaction time is roughly $0.2\,\mathrm{s}$, and it lands on the total you measure however long that total is. Time one oscillation and that $0.2\,\mathrm{s}$ is an error of perhaps 30%. Time ten and you divide the *same* $0.2\,\mathrm{s}$ by ten before it reaches T — an uncertainty of about $0.02\,\mathrm{s}$.

You have not made your reactions faster. You have made them matter ten times less. This trick is worth remembering: it is one of the few places in physics where you get accuracy for free.

Finding your uncertainties

Two measurements today, so two uncertainties. Work both out before you start.

Step 1 — the masses

The blocks say $100\,\mathrm{g}$. Nobody has checked. Weigh **ten** of them.

Block 1	**2**	**3**	**4**	**5**
Block 6	**7**	**8**	**9**	**10**

Largest ________g Smallest ________g Range ________g $\Delta m = \frac{\text{range}}{2} =$ ________g

Better still: weigh your own load, hanger included, and use that instead of the label.

Step 2 — the timing

The stopwatch reads to $0.01\,\mathrm{s}$, but *you* have to stop it. Start it and try to stop it at exactly $5.00\,\mathrm{s}$, without watching the display. Record what you got. **Six tries.**

Try 1	**2**	**3**	**4**	**5**	**6**

Largest ________s Smallest ________s Range ________s $\Delta t = \frac{\text{range}}{2} =$ ________s

Ten oscillations, so one period is ten times better: $\Delta T = \Delta t/10 =$ __________s

Step 3 — commit to a prediction

Say what you expect, before the data can tell you

A prediction you got wrong and then understood is worth more than one you got right by accident. Answer these now, in pen, and do not change them afterwards — the point is to be able to compare.

1. Will a heavier mass oscillate *faster* or *slower*? Why do you think so?

2. Sketch the shape you expect for a graph of T against m. Straight line? Curve? Through the origin?

3. Will pulling the mass down further change the period?

Results

Record every reading as you take it. Use the Δm and ΔT you worked out on the previous page — an uncertainty invented afterwards is a guess, and the IB is very good at spotting guesses.

Mass m / kg	**Time for 10 oscillations** t / s			**Mean** t / s	**Period** T / s	T^2 / s^2
	Trial 1	**Trial 2**	**Trial 3**			

The amplitude check. Keep one mass on the spring and change only how far you pull it down.

Amplitude / cm	t **for 10** / s	**Period** T / s

What do these three periods tell you?

Processing

1. Plot a graph of T^2 (vertical) against m (horizontal).
2. Draw the best straight line through your points.
3. Find its gradient.

Why T^2 against m, and not T against m

Square both sides of the prediction:

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad \Longrightarrow \qquad T^2 = \frac{4\pi^2}{k}\,m$$

That is now $y = mx$ — a straight line, with

$$\text{gradient} = \frac{4\pi^2}{k} \qquad \text{so} \qquad k = \frac{4\pi^2}{\text{gradient}}$$

A straight line is the only shape the eye can judge honestly. Whenever you can turn a curve into a line by choosing what to plot, do it. The IB calls this **linearizing**, and it is a named skill in the guide.

Expect a small positive intercept, and do not treat it as a mistake. In theory the line goes through the origin. In practice the spring has mass too, so there is already something oscillating when your hanging mass is zero. The gradient — and therefore k — is unaffected.

My gradient = ________________ s^2/kg **so** k **=** ________________ N/m

Next steps

1. **Before you leave:** table complete, Δm and ΔT written down.
2. **For Tuesday:** work out T^2, plot it against m in a spreadsheet with **uncertainty bars** on T^2, and bring it — we run the lesson from your graphs. Vertical bars only: you weighed your masses, so Δm is smaller than the dot. Say so on the graph.